Chinese *gèng* Expresses Implicit Comparison

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Overview

- Chinese does not make a morphological distinction between the positive and comparative use of a gradable adjective.
- In Chinese, the particle *gèng* often appears in constructions of comparison. What is its meaning contribution?
- We propose that:

- *Gèng*-sentences express implicit comparison (not explicit comparison), involving a positive use (not comparative use) of gradable adjectives;
- *Gèng* semantically introduces a new positive threshold between the items (target & standard) under comparison;
- For a *gèng*-sentence, the evaluative inference that the standard already reaches the regular positive threshold is an implicature, not entailed meaning (cf. the semantics of *even*).

Data

- Chinese lacks an *er*-like morpheme. (1) is ambiguous between a positive and a comparative reading:

(1) John hé Mary, shéi gāo?
and who tall(er)
✓Positive: 'Who is tall, John and Mary?'
✓Comparative: 'Who is taller, John and Mary?'

- *Gèng* can optionally appear, guaranteeing a seemingly comparative meaning. This is actually an implicit comparison.

(2) John hé Mary, shéi gèng gāo?
and who GENG tall(er)
#Positive: 'Who is tall, John and Mary?'
✓Comparative: 'Who is taller, John and Mary?'

- **Evidence 1**: Incompatible with numerical differentials.

(3) John bǐ Mary gèng gāo (*wǔ límǐ).
STDD GENG tall(er) 5 cm
'Compared to Mary, John is (*5 cm) tall.'

Abbreviations: 1 = first person, NEG = negative, PFV = perfective, PRT = particle, SG = singular, STDD = standard marker.
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Data (Cont'd)

- **Evidence 2**: Incompatible with 'crisp judgment' context.
- (5)'s infelicity under (4) argues against the view that *gèng* works in the same way as *even* (pace Liu 2010, Chen 2023). (6) is felicitous under this 'crisp judgment' context.

(4) Context: John is 6' tall; Mary is 5'11" tall. Both are tall.
(5) John bǐ Mary gèng gāo.
STDD GENG tall
Compared to Mary, John is tall. [Infelicitous under (4)]
(6) John is even taller than Mary. [Felicitous under (4)]

- **Evidence 3**: Similar to implicit comparison (but not explicit comparison, see Kennedy 2007, Sawada 2009), a comparison with the use of *gèng* has evaluative inferences.

(7) Evaluative inference of (5)/(6): Mary is already tall.

- (8, 9) show that this kind of evaluative inference for a *gèng*-sentence can be suspended or cancelled, suggesting that it's an implicature (cf. for (6), this inference is a presupposition, see Zhang 2022).

(8) Wǒ bù zhīdào Mary gāo-bù-gāo, dàn John kěndìng
1SG NEG know tall-NEG-tall but John definitely
bǐ Mary gèng gāo.
STDD GENG tall
'Though I don't know whether Mary is tall, compared to her, John is definitely tall.' ['Mary is tall' suspended]
(9) Mary gàn-de bù hǎo, dàn yě méi rén gàn-de bǐ
do-PRT not well but still NEG person do-PRT STDD
Mary gèng hǎo-le.
GENG well-PFV
'Mary didn't do well, but still, compared to her, no one did well.' ['Mary did well' explicitly cancelled]

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Analysis

- Following Zhang & Zhang-Yukun (2024), we argue that gradable adjectives encode inequality *per se*.
- A gradable adjective takes two degree arguments & an individual argument, and returns an update function (10).
- E.g. variable i_{tall} maps to the positive threshold $d_{\text{tall-pos}}$ (11).
- In comparatives, the reference is the measurement of the standard, and the difference can be accommodated (12) or numerically specified (13).

(10) $[\![\text{gāo}]\!] = \lambda d_{\text{DIFF}}.\lambda d_{\text{STDD}}.\lambda x.\lambda C.[d_{\text{DIFF}} > 0].$
$\{\langle w, g\rangle \in C \mid \text{HEIGHT}_w(x) \geq d_{\text{STDD}} + d_{\text{DIFF}}\}.$
(11) $[\![\text{POS}^{i_{\text{tall}}}\ \text{gāo}]\!] = \lambda x.\lambda C.[d_{\text{DIFF}} > 0].$
$\{\langle w, g\rangle \in C \mid \text{HEIGHT}_w(x) \geq d_{\text{tall-pos}} + d_{\text{DIFF}}\}.$
(12) $[\![\text{bǐ Mary gāo}]\!] = \lambda x.\lambda C.[d_{\text{DIFF}} > 0].$
$\{\langle w, g\rangle \in C \mid \text{HEIGHT}_w(x) \geq \text{HEIGHT}_w(m) + d_{\text{DIFF}}\}.$
(13) $[\![\text{bǐ Mary gāo 5cm}]\!] = \lambda x.\lambda C.$
$\{\langle w, g\rangle \in C \mid \text{HEIGHT}_w(x) \geq \text{HEIGHT}_w(m) + 5\text{cm}\}.$

- A *gèng*-sentence is an implicit comparison, which involves the positive use of the adjective.
- *Gèng* (i) introduces **a degree dref** which maps to a new threshold, and (ii) restricts it to a degree value above the standard.

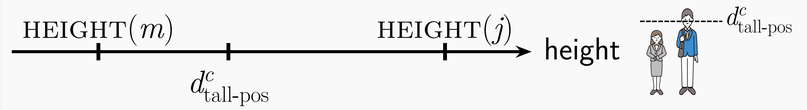


Figure: [John bǐ Mary gèng gāo]: Compared to M., J. is tall.

(14) $[\![\text{gèng}^{i'_{\text{tall}}}]\!] = \lambda\mathcal{G}.\lambda d_{\text{STDD}}.\lambda x.\lambda C.\{\langle w, g^{i'_{\text{tall}} \mapsto d^c_{\text{tall-pos}}}\rangle \mid \langle w, g\rangle \in \mathcal{G}(d_{\text{DIFF}})(d^c_{\text{tall-pos}})(x)(C),\ d^c_{\text{tall-pos}} > d_{\text{STDD}},\ d_{\text{DIFF}} > 0\}.$
(15) $[\![\text{gèng}^{i'_{\text{tall}}}\text{gāo}]\!] = \lambda d_{\text{STDD}}.\lambda x.\lambda C.\{\langle w, g^{i'_{\text{tall}} \mapsto d^c_{\text{tall-pos}}}\rangle \mid \langle w, g\rangle \in C, \text{HT}_w(x) \geq d^c_{\text{tall-pos}} + d_{\text{DIFF}},\ d^c_{\text{tall-pos}} > d_{\text{STDD}},\ d_{\text{DIFF}} > 0\}.$
(16) $[\![\text{(bǐ M.) gèng}^{i'_{\text{tall}}}\text{gāo}]\!] = \lambda x.\lambda C.\{\langle w, g^{i'_{\text{tall}} \mapsto d^c_{\text{tall-pos}}}\rangle \mid \langle w, g\rangle \in C, \text{HT}_w(x) \geq d^c_{\text{tall-pos}} + d_{\text{DIFF}},\ d^c_{\text{tall-pos}} > \text{HT}_w(m),\ d_{\text{DIFF}} > 0\}.$

- Note that using *gèng* does not change the regular threshold.

Selected References

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